

MELBOURNE LAVERTON, AustraliaWMO: **948650**Lat: **37.8564S**Lon: **144.7567E**Elev: **67**StdP: **14.66**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.3	37.3	31.5	25.9	52.1	33.4	28.0	49.3	27.9	55.1	25.4	55.0	6.2	310	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.5	95.7	66.3	90.5	65.3	85.4	64.6	70.1	83.1	68.4	81.7	67.0	79.7	15.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	97.8	73.6	64.6	91.3	72.2	62.9	86.1	70.9	34.1	83.1	32.7	81.6	31.5	79.9	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.6	22.8	20.4	DB	30.8	105.5	1.9	4.5	29.5	108.7	28.4	111.4	27.4	113.9	26.0	117.2	(4)
(5)			WB	30.2	73.4	1.8	1.6	28.9	74.6	27.8	75.5	26.8	76.3	25.4	77.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.3	68.0	68.2	64.6	58.9	54.4	50.4	49.6	50.5	53.5	57.1	61.1	64.3	(6)	
	DBStd	8.62	7.04	6.46	6.24	5.44	4.68	3.88	3.69	4.15	5.15	6.10	6.53	6.91	(7)	
	HDD50	168	0	0	0	1	9	41	54	43	16	4	0	0	(8)	
	HDD65	2912	40	32	83	198	330	438	479	449	349	260	156	98	(9)	
	CDD50	3207	557	509	454	269	145	53	40	59	121	224	334	442	(10)	
	CDD65	474	132	121	72	15	1	0	0	0	3	15	40	75	(11)	
	CDH74	5276	1394	1152	734	168	5	0	0	1	29	271	544	978	(12)	
	CDH80	2420	729	533	308	39	0	0	0	0	4	89	219	499	(13)	
(14) Wind	WSAvg	10.4	11.1	10.7	10.2	9.2	9.5	9.3	10.4	10.7	11.1	11.1	10.9	11.1	(14)	
(15) Precipitation	PrecAvg	21.5	1.8	1.6	1.2	1.9	1.8	1.5	1.5	1.8	1.9	2.1	2.1	1.9	(15)	
	PrecMax	31.9	5.6	10.7	4.8	7.9	4.8	3.8	3.2	3.5	5.0	6.5	5.2	4.5	(16)	
	PrecMin	11.6	0.1	0.0	0.2	0.3	0.2	0.4	0.4	0.4	0.6	0.2	0.2	0.1	(17)	
	PrecStd	5.0	1.3	2.0	0.8	1.6	1.1	0.7	0.7	0.7	0.9	1.2	1.2	1.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	104.8	98.8	96.1	84.8	73.6	64.5	64.1	69.5	78.4	89.0	92.9	100.4	(19)	
		MCWB	67.9	66.9	66.5	61.8	57.4	54.7	53.4	55.9	58.6	61.1	64.5	66.8	(20)	
	2%	DB	95.6	92.7	88.2	78.7	68.9	61.8	60.6	64.6	71.6	81.3	86.3	92.8	(21)	
		MCWB	66.8	67.0	64.9	60.1	57.2	54.0	52.1	53.8	57.1	60.3	63.6	64.8	(22)	
	5%	DB	88.3	86.9	82.3	74.0	65.8	59.6	58.4	61.6	67.6	75.5	80.2	85.1	(23)	
		MCWB	66.1	66.2	63.9	59.5	56.0	52.9	50.8	52.2	55.6	58.6	62.5	63.8	(24)	
	10%	DB	80.9	80.8	76.6	69.6	63.0	57.6	56.5	58.9	64.1	70.1	74.4	77.6	(25)	
		MCWB	65.7	65.7	63.1	58.6	55.0	51.9	50.1	50.8	54.0	56.8	60.8	63.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.1	71.9	69.8	64.6	61.0	57.4	55.2	57.3	61.3	64.8	69.1	69.9	(27)	
		MCDB	85.8	86.1	85.2	76.9	67.9	61.7	60.9	66.5	71.9	78.2	83.0	83.0	(28)	
	2%	WB	70.4	69.8	67.4	62.7	58.7	55.3	53.3	54.8	58.5	62.0	66.2	67.4	(29)	
		MCDB	83.8	81.6	79.8	71.7	65.7	59.7	58.6	62.9	68.4	75.0	79.0	82.3	(30)	
	5%	WB	68.4	68.2	65.9	61.2	57.1	53.7	51.9	53.0	56.6	59.8	64.1	65.7	(31)	
		MCDB	82.5	80.2	76.8	70.0	63.8	58.3	56.9	59.7	66.1	72.0	76.1	79.5	(32)	
	10%	WB	66.5	66.7	64.2	59.5	55.7	52.5	50.7	51.6	54.6	57.8	62.0	63.8	(33)	
		MCDB	79.7	78.0	74.6	68.0	61.9	56.9	55.5	57.7	62.8	68.3	72.7	76.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	18.5	18.1	17.7	15.1	13.6	13.7	15.5	17.9	19.7	19.0	19.7	(35)	
		MCDBR	34.9	32.1	30.3	26.1	18.8	15.7	15.4	19.3	23.6	30.8	32.1	35.7	(36)	
	5% WB	MCWBR	11.1	10.8	11.3	11.6	10.1	9.8	9.2	10.6	12.1	13.4	13.0	12.3	(37)	
		MCWBR	26.2	23.8	22.7	20.6	16.4	13.9	13.8	17.8	21.3	26.0	26.0	26.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.352	0.357	0.331	0.332	0.315	0.302	0.303	0.316	0.333	0.338	0.349	0.344	(40)	
		taud	2.505	2.499	2.573	2.537	2.569	2.593	2.581	2.530	2.483	2.486	2.482	2.514	(41)	
	Ebn at Noon		311	302	298	277	263	258	265	278	292	305	310	315	(42)	
		Edh at Noon	36	34	30	27	23	20	22	26	31	34	36	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		2267	1980	1574	1094	727	566	633	894	1277	1702	1995	2239	(44)	
	RadStd		142	100	90	63	41	43	36	57	88	132	119	127	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+1.08	+0.65	+0.97	N/A	+0.72	+0.85	-52	-272	+358	+130	(46)
(47) Regional (38 neighbors)		+0.92	N/A	N/A	+1.57	N/A	N/A	-31	-232	+315	+106	(47)

Nomenclature: See separate page